set the intellectual tone of the subject. And, while a few others were willing to take black holes seriously, Einstein and Eddington were not. Black holes just didn't "smell right"; they were outrageously bizarre; they violated Einstein's and Eddington's intuitions about how our Universe ought to behave.

In the 1920s Einstein seems to have dealt with the issue by ignoring it. Nobody was pushing black holes as a serious prediction, so there was not much need on that score to straighten things out. And since other mysteries of nature were more interesting and puzzling to Einstein, he put his energies elsewhere.

Eddington in the 1920s took a more whimsical approach. He was a bit of a ham, he enjoyed popularizing science, and so long as nobody was taking black holes too seriously, they were a playful thing to dangle in front of others. Thus, we find him writing in 1926 in his book *The Internal Constitution of the Stars* that no observable star can possibly be more compact than the critical circumference: "Firstly," he wrote, "the force of gravitation would be so great that light would be unable to escape from it, the rays falling back to the star like a stone to the Earth. Secondly, the redshift of the spectral lines would be so great that the spectrum would be shifted out of existence. Thirdly, the mass would produce so much curvature of the space-time metric that space would close up round the star, leaving us outside (i.e. nowhere)." The first conclusion was the Newtonian version of light not escaping; the second was a semi-accurate, relativistic description; and the third was typical Eddingtonian hyperbole. As one sees clearly from the embedding diagrams of Figure 3.4, when a star is as small as the critical circumference, the curvature of space is strong but not infinite, and space is definitely not wrapped up around the star. Eddington may have known this, but his description made a good story, and it captured in a whimsical way the spirit of Schwarzschild's spacetime curvature.

In the 1930s, as we shall see in Chapter 4, the pressure to take black holes seriously began to mount. As the pressure mounted, Eddington, Einstein, and others among the "opinion setters" began to express unequivocal opposition to these outrageous objects.

In 1939, Einstein published a general relativistic calculation that he interpreted as an example of why black holes cannot exist. His calculation analyzed the behavior of an idealized kind of object which one might have thought could be used to make a black hole. The object was a cluster of